

## PLATTSBURGH, NY, USA

WMO: 726225

Lat: 44.650N

Lon: 73.467W

Elev: 234

StdP: 14.57

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 64776

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -8.8       | -3.2 | -18.2                         | 2.1 | -6.3 | -13.2 | 2.7 | -1.4 | 22.9                  | 27.1 | 20.3 | 23.7 | 4.6                   | 280  | 0.531 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.7                   | 87.6            | 71.6 | 83.9 | 70.4 | 81.3 | 68.6 | 74.4                | 83.0 | 72.6 | 80.4 | 70.9 | 78.2 | 8.3                  | 130  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 71.9                            | 118.8 | 79.4 | 70.1 | 111.7 | 77.4 | 68.3 | 104.7 | 75.5 | 38.1          | 83.5 | 36.4 | 80.3 | 34.9 | 78.0 | 82.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |     |
| (4) 19.8          | 17.6 | 15.7 | DB                         | -16.4 | 93.6               | 7.2 | 2.9                                                | -21.6 | 95.7       | -25.8 | 97.4       | -29.8 | 99.0       | -35.1 | 101.1 | (4) |
| (5)               |      |      | WB                         | -16.8 | 77.7               | 7.0 | 2.0                                                | -21.8 | 79.1       | -25.9 | 80.3       | -29.9 | 81.5       | -35.0 | 83.0  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 45.4   | 18.5  | 20.7  | 30.3  | 43.7  | 56.1  | 64.6  | 69.8  | 67.9  | 60.5  | 48.3  | 37.4  | 25.9  |  |  |
|                                                                        | DBStd       | 19.88  | 13.01 | 11.29 | 10.45 | 8.39  | 7.32  | 6.56  | 5.10  | 5.22  | 7.09  | 7.91  | 8.63  | 10.65 |  |  |
|                                                                        | HDD50       | 3927   | 976   | 820   | 614   | 226   | 23    | 0     | 0     | 0     | 5     | 131   | 384   | 748   |  |  |
|                                                                        | HDD65       | 7549   | 1441  | 1240  | 1076  | 641   | 290   | 86    | 13    | 30    | 172   | 520   | 827   | 1213  |  |  |
|                                                                        | CDD50       | 2262   | 0     | 0     | 3     | 37    | 213   | 437   | 614   | 554   | 319   | 77    | 7     | 1     |  |  |
|                                                                        | CDD65       | 407    | 0     | 0     | 0     | 2     | 15    | 73    | 161   | 119   | 36    | 1     | 0     | 0     |  |  |
|                                                                        | CDH74       | 3408   | 0     | 0     | 1     | 45    | 201   | 674   | 1320  | 869   | 289   | 9     | 0     | 0     |  |  |
|                                                                        | CDH80       | 862    | 0     | 0     | 0     | 10    | 50    | 191   | 359   | 191   | 61    | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 6.1    | 6.3   | 6.5   | 7.4   | 7.6   | 6.6   | 5.8   | 5.1   | 4.9   | 5.3   | 5.7   | 6.0   | 5.7   |  |  |
|                                                                        | PrecAvg     | 26.5   | 1.6   | 1.3   | 1.7   | 2.4   | 2.4   | 2.5   | 2.5   | 3.3   | 2.6   | 2.2   | 2.4   | 1.9   |  |  |
| (15) Precipitation                                                     | PrecMax     | 42.0   | 6.5   | 3.2   | 3.0   | 6.4   | 5.9   | 6.2   | 4.7   | 6.5   | 6.8   | 4.8   | 4.9   | 4.9   |  |  |
|                                                                        | PrecMin     | 17.1   | 0.3   | 0.4   | 0.2   | 0.4   | 0.8   | 0.7   | 0.9   | 0.8   | 0.6   | 0.2   | 0.5   | 0.3   |  |  |
|                                                                        | PrecStd     | 6.6    | 1.3   | 0.7   | 0.7   | 1.1   | 1.1   | 1.3   | 1.0   | 1.3   | 1.3   | 1.0   | 1.1   | 1.1   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 51.5  | 51.9  | 64.4  | 80.2  | 86.8  | 90.8  | 91.4  | 89.6  | 86.2  | 74.8  | 64.5  | 53.8  |  |  |
|                                                                        |             | MCWB   | 47.6  | 44.4  | 52.3  | 63.0  | 66.2  | 72.1  | 74.3  | 73.4  | 71.8  | 63.5  | 56.4  | 48.3  |  |  |
|                                                                        | 2%          | DB     | 44.2  | 44.9  | 55.1  | 71.6  | 79.3  | 85.2  | 87.7  | 84.4  | 81.1  | 69.9  | 59.4  | 48.1  |  |  |
|                                                                        |             | MCWB   | 40.1  | 39.0  | 45.7  | 55.9  | 63.2  | 70.2  | 72.3  | 70.8  | 68.9  | 61.2  | 53.4  | 44.2  |  |  |
|                                                                        | 5%          | DB     | 40.1  | 40.1  | 49.3  | 64.3  | 74.7  | 81.2  | 83.9  | 81.5  | 77.2  | 65.5  | 54.9  | 43.9  |  |  |
|                                                                        |             | MCWB   | 36.9  | 35.6  | 41.7  | 51.4  | 60.6  | 67.8  | 71.1  | 69.2  | 67.8  | 58.1  | 49.1  | 40.1  |  |  |
|                                                                        | 10%         | DB     | 36.6  | 36.6  | 44.9  | 57.9  | 70.5  | 77.5  | 81.2  | 79.0  | 73.1  | 61.3  | 51.6  | 39.7  |  |  |
|                                                                        |             | MCWB   | 33.7  | 33.2  | 39.0  | 47.4  | 58.9  | 65.2  | 69.3  | 68.0  | 64.5  | 55.0  | 46.6  | 36.6  |  |  |
|                                                                        | 0.4%        | WB     | 48.5  | 45.1  | 54.3  | 64.4  | 69.9  | 75.3  | 77.8  | 75.7  | 73.9  | 66.5  | 58.8  | 49.9  |  |  |
|                                                                        |             | MCDB   | 51.1  | 49.5  | 62.2  | 77.1  | 79.2  | 84.9  | 87.8  | 84.9  | 82.0  | 72.0  | 63.1  | 52.4  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 41.0  | 39.7  | 47.3  | 58.0  | 66.7  | 72.6  | 75.0  | 73.4  | 71.1  | 62.5  | 54.0  | 44.7  |  |  |
|                                                                        |             | MCDB   | 43.3  | 43.9  | 53.7  | 68.2  | 75.5  | 82.0  | 83.7  | 80.9  | 78.5  | 67.9  | 58.4  | 47.4  |  |  |
|                                                                        | 5%          | WB     | 37.1  | 36.3  | 42.5  | 52.8  | 63.5  | 70.0  | 73.0  | 71.5  | 68.7  | 58.8  | 50.2  | 40.8  |  |  |
|                                                                        |             | MCDB   | 39.8  | 39.7  | 48.1  | 61.4  | 70.4  | 78.1  | 80.5  | 78.2  | 75.2  | 64.6  | 54.3  | 43.7  |  |  |
|                                                                        | 10%         | WB     | 33.7  | 33.3  | 39.5  | 49.0  | 60.4  | 67.6  | 71.2  | 69.9  | 66.2  | 55.8  | 46.5  | 37.0  |  |  |
|                                                                        |             | MCDB   | 36.4  | 36.2  | 44.7  | 57.2  | 68.3  | 73.9  | 78.5  | 76.3  | 72.2  | 60.1  | 50.4  | 39.6  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 16.6  | 17.8  | 17.4  | 19.3  | 20.8  | 19.8  | 19.7  | 19.9  | 20.4  | 17.6  | 15.6  | 14.2  |  |  |
|                                                                        |             | MCDBR  | 16.9  | 19.1  | 24.7  | 30.8  | 28.6  | 25.2  | 22.9  | 22.8  | 24.3  | 22.6  | 22.0  | 15.5  |  |  |
|                                                                        |             | MCWBR  | 14.8  | 15.4  | 16.9  | 18.1  | 14.9  | 12.4  | 11.3  | 11.6  | 13.8  | 14.6  | 17.2  | 13.5  |  |  |
|                                                                        |             | MCDBR  | 16.9  | 18.7  | 22.9  | 28.4  | 23.2  | 22.0  | 20.1  | 20.0  | 21.5  | 21.2  | 20.3  | 15.6  |  |  |
|                                                                        | 5% WB       | MCWBR  | 15.1  | 15.7  | 16.7  | 18.3  | 14.0  | 12.3  | 11.3  | 11.3  | 13.1  | 14.9  | 17.3  | 14.2  |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.286 | 0.289 | 0.329 | 0.353 | 0.379 | 0.438 | 0.440 | 0.417 | 0.372 | 0.337 | 0.312 | 0.283 |  |  |
|                                                                        | taud        |        | 2.345 | 2.342 | 2.364 | 2.399 | 2.382 | 2.220 | 2.220 | 2.277 | 2.395 | 2.495 | 2.491 | 2.410 |  |  |
|                                                                        | Ebn at Noon |        | 264   | 286   | 287   | 288   | 283   | 265   | 263   | 265   | 269   | 266   | 254   | 254   |  |  |
|                                                                        | Edh at Noon |        | 24    | 30    | 34    | 36    | 37    | 44    | 43    | 40    | 32    | 26    | 22    | 21    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 457   | 726   | 1090  | 1372  | 1602  | 1729  | 1764  | 1580  | 1277  | 746   | 470   | 334   |  |  |
|                                                                        | RadStd      |        | 49    | 69    | 79    | 167   | 174   | 149   | 135   | 72    | 102   | 77    | 55    | 28    |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (48 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | +113  | N/A   |       |

Nomenclature: See separate page